Forecasting U.S. Gasoline Prices under Geopolitical Shocks

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1. Project Overview

We want to analyze how major geopolitical shocks affect U.S. gasoline prices and compare different time series models to see which performs best under these conditions.

- **Russia–Ukraine War** (Feb 2022)
- **Israel–Iran Conflict** (June 2025)
- **US Intervention in Venezuela** (Jan 2026)

2. Data Loading & Preprocessing

To analyze the relationship between U.S. gasoline prices, crude oil prices, and gasoline stocks, we collected weekly data from the U.S. Energy Information Administration (EIA).

During the initial data loading phase, we identified a structural reporting discrepancy: weekly gasoline prices are recorded on **Mondays**, whereas WTI crude oil prices and gasoline stocks are reported on **Fridays**. To align the datasets correctly for our time series join, we advanced the WTI and stocks dates by 3 days.

After successfully merging the data and filtering the timeline to start from 2018, we applied a rigorous data cleaning process. We utilized linear interpolation to impute any missing values and applied STL decomposition-based smoothing to handle extreme outliers. This ensures that our forecasting models are robust and not skewed by temporary noise.

```r
# Load required packages
library(tidyverse)
library(lubridate)
library(timetk)
```

```r
# Guys, Please type install.packages(c("tidymodels", "modeltime", "prophet", "forecast", "xgb

# ==========================================
# 1. Read Data and Align Dates
# ==========================================
# Note: EIA CSV files contain 4 lines of metadata at the top, so we use skip = 4.

# Gas prices are reported on Mondays.
gas_price_df <- read_csv("Raw_Data/Weekly U.S. Gas Price (EIA)", skip = 4) %>%
  rename(date = 1, gas_price = 2) %>%
  mutate(date = mdy(date))

# WTI prices and Gas stocks are reported on Fridays.
# We add 3 days to align them with the Monday schedule of Gas prices.
wti_price_df <- read_csv("Raw_Data/Weekly WTI Price (EIA)", skip = 4) %>%
  rename(date = 1, wti_price = 2) %>%
  mutate(date = mdy(date) + days(3))

gas_stocks_df <- read_csv("Raw_Data/Weekly U.S. Gas Stocks (EIA)", skip = 4) %>%
  rename(date = 1, gas_stocks = 2) %>%
  mutate(date = mdy(date) + days(3))

# ==========================================
# 2. Merge and Clean Data
# ==========================================
clean_data <- gas_price_df %>%
  left_join(wti_price_df, by = "date") %>%
  left_join(gas_stocks_df, by = "date") %>%
  arrange(date) %>% # Ensure chronological order
  filter(date >= as.Date("2018-01-01")) %>%
  mutate(
    # Step 1: Impute missing values (NAs) using interpolation
    # Step 2: Clean and smooth extreme anomalies out of the time series
    gas_price  = ts_clean_vec(ts_impute_vec(gas_price, period = 1), period = 1),
    wti_price  = ts_clean_vec(ts_impute_vec(wti_price, period = 1), period = 1),
    gas_stocks = ts_clean_vec(ts_impute_vec(gas_stocks, period = 1), period = 1)
  )

# Preview the cleaned and aligned data
head(clean_data)
```

```
# A tibble: 6 x 4
```

```
  date       gas_price wti_price gas_stocks
  <date>         <dbl>     <dbl>      <dbl>
1 2018-01-01      2.52      59.9      24987
2 2018-01-08      2.52      61.4      24375
3 2018-01-15      2.56      63.3      25124
4 2018-01-22      2.57      63.8      24019
5 2018-01-29      2.61      65.1      23449
6 2018-02-05      2.64      65.3      24511
```

```r
# ==========================================
# 3. Export Processed Data
# ==========================================
# We save the cleaned and merged dataset for reproducibility and team sharing.
# The file is saved in CSV format within the 'Processed_Data' folder.

write_csv(clean_data, "Processed_Data/cleaned_data.csv")
```

3. Exploratory Data Analysis (EDA)

And we visualize the cleaned gasoline prices.

```r
# Plot the target variable
clean_data %>%
  plot_time_series(date, gas_price,
                   .interactive = TRUE,
                   .title = "Cleaned U.S. Gasoline Prices")
```

Cleaned U.S. Gasoline Prices
4.5
4
3.5
3
2.5
2
2018
2019
2020
2021
2022
2023

4. Feature Engineering & Train/Test Split

We will create dummy variables to represent our three major geopolitical shocks. Then, we will split the data, reserving the most recent 10% (approx. 26 weeks) for model testing.

```
# Define shock dates
shock_ukraine <- as.Date("2022-02-24")
shock_israel_iran <- as.Date("2025-06-01") # June 2025
shock_venezuela <- as.Date("2026-01-01")   # Jan 2026

# Add dummy variables to the cleaned data
model_data <- clean_data %>%
  mutate(
    shock_ukraine_dummy = ifelse(date >= shock_ukraine, 1, 0),
    shock_israel_iran_dummy = ifelse(date >= shock_israel_iran, 1, 0),
    shock_venezuela_dummy = ifelse(date >= shock_venezuela, 1, 0)
  )

# Time-based split: 90% Training, 10% Testing
library(tidymodels)
library(modeltime)

splits <- initial_time_split(model_data, prop = 0.9)
```

5. Expanded Modeling Setup

To thoroughly evaluate the impact of geopolitical shocks, we expand our forecasting roster to include 7 distinct models. We structure our approach logically, progressing from a simple baseline to traditional statistical models, machine learning algorithms, and finally, complex neural networks.

```
# 1. Seasonal Naive (Simplest Baseline)
model_snaive <- naive_reg() %>%
  set_engine("snaive") %>%
  fit(gas_price ~ date, data = training(splits))

# 2. ETS (Error, Trend, Seasonality)
model_ets <- exp_smoothing() %>%
  set_engine("ets") %>%
  fit(gas_price ~ date, data = training(splits))
```

```r
# 3. ARIMAX (Traditional Statistical Model)
model_arima <- arima_reg() %>%
  set_engine("auto_arima") %>%
  fit(gas_price ~ date + wti_price + gas_stocks + shock_ukraine_dummy + shock_israel_iran_du
      data = training(splits))

# 4. Prophet (Generalized Additive Model)
model_prophet <- prophet_reg() %>%
  set_engine("prophet") %>%
  fit(gas_price ~ date + wti_price + gas_stocks + shock_ukraine_dummy + shock_israel_iran_du
      data = training(splits))

# 5. Random Forest (Machine Learning)
# Added .set_mode("regression") to specify we are predicting continuous values
model_rf <- rand_forest(trees = 500, min_n = 5) %>%
  set_engine("ranger") %>%
  set_mode("regression") %>%
  fit(gas_price ~ date + wti_price + gas_stocks + shock_ukraine_dummy + shock_israel_iran_du
      data = training(splits))

# 6. ARIMA-Boost (Hybrid/Ensemble Model)
# Added .set_mode("regression") for the XGBoost component
model_arima_boost <- arima_boost() %>%
  set_engine("auto_arima_xgboost") %>%
  set_mode("regression") %>%
  fit(gas_price ~ date + wti_price + gas_stocks + shock_ukraine_dummy + shock_israel_iran_du
      data = training(splits))

# 7. NNAR (Neural Network Auto-Regression)
model_nnar <- nnetar_reg() %>%
  set_engine("nnetar") %>%
  fit(gas_price ~ date + wti_price + gas_stocks + shock_ukraine_dummy + shock_israel_iran_du
      data = training(splits))
```

6. Comprehensive Model Evaluation

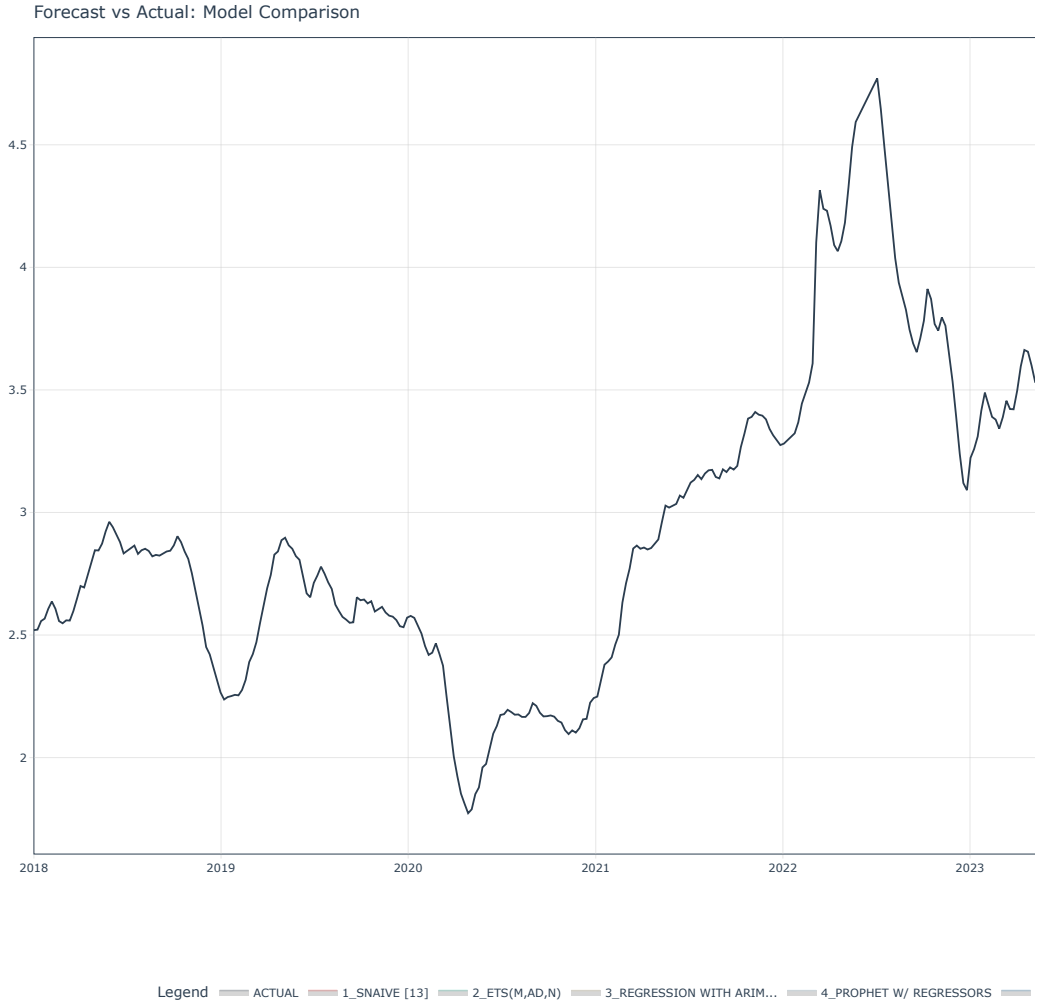
We consolidate all 7 models into a single table to benchmark their forecasting accuracy (e.g., RMSE, MAE, MAPE) on the withheld test dataset.

```r
# Create a Modeltime Table maintaining the simple-to-complex order
models_tbl <- modeltime_table(
```

```r
  model_snaive,
  model_ets,
  model_arima,
  model_prophet,
  model_rf,
  model_arima_boost,
  model_nnar
)

# Calibrate models on the testing data
calibration_tbl <- models_tbl %>%
  modeltime_calibrate(new_data = testing(splits))

# Extract and display the accuracy metrics (Sorted by RMSE)
accuracy_tbl <- calibration_tbl %>%
  modeltime_accuracy() %>%
  arrange(rmse)

print(accuracy_tbl)
```

```
# A tibble: 7 x 9
  .model_id .model_desc              .type   mae  mape  mase smape  rmse     rsq
      <int> <chr>                    <chr> <dbl> <dbl> <dbl> <dbl> <dbl>   <dbl>
1         4 PROPHET W/ REGRESSORS    Test  0.122  3.87  2.55  3.94 0.135 0.861
2         5 RANGER                   Test  0.163  5.26  3.42  5.15 0.211 0.806
3         3 REGRESSION WITH ARIMA(1~ Test  0.170  5.42  3.55  5.33 0.229 0.779
4         7 NNAR(1,1,10)[13]         Test  0.169  5.24  3.53  5.26 0.252 0.863
5         6 ARIMA(1,1,0)(0,0,1)[13]~ Test  0.178  5.35  3.73  5.54 0.288 0.0259
6         1 SNAIVE [13]              Test  0.199  6.15  4.16  6.21 0.290 0.00124
7         2 ETS(M,AD,N)              Test  0.178  5.30  3.72  5.54 0.296 0.00102
```

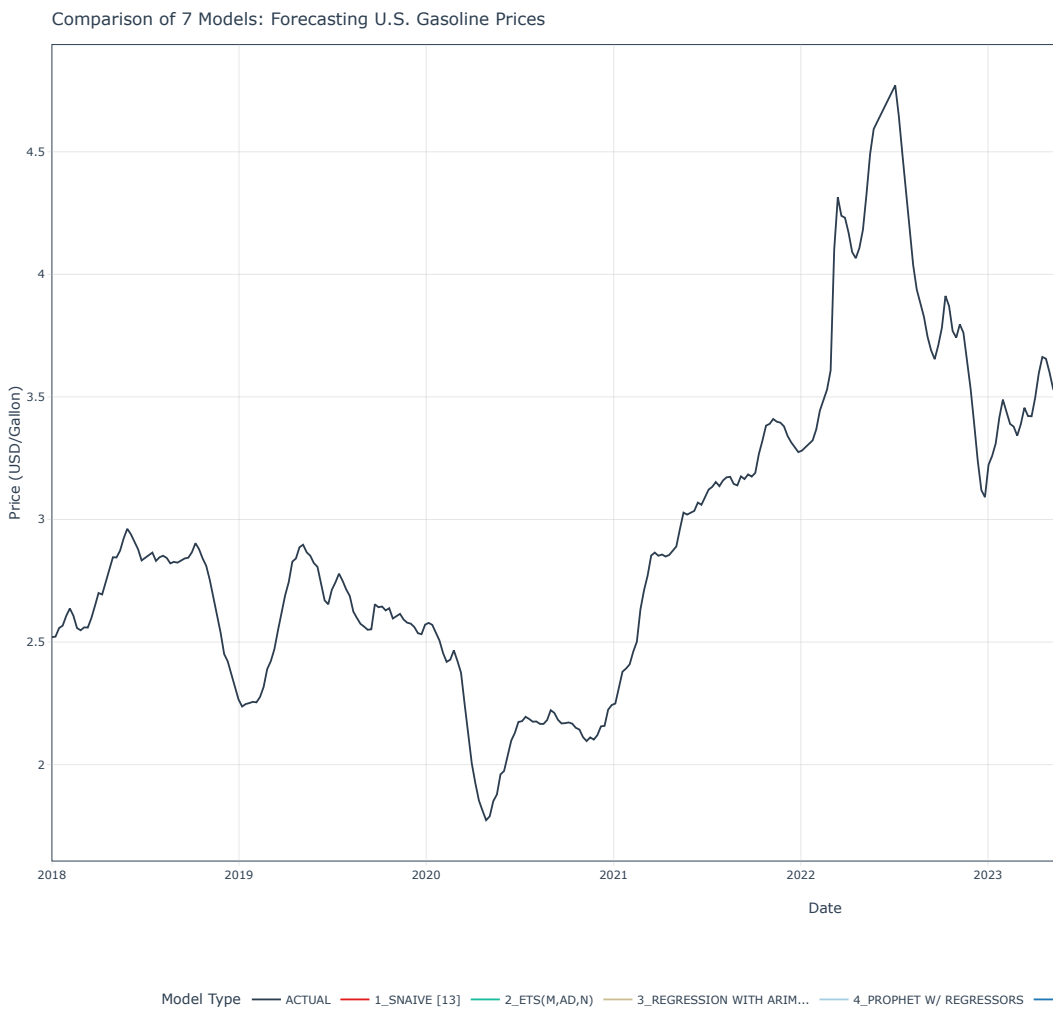
```r
# Plot Forecasts vs Actuals for all models
calibration_tbl %>%
  modeltime_forecast(
    new_data    = testing(splits),
    actual_data = clean_data
  ) %>%
  plot_modeltime_forecast(
    .legend_max_width = 25,
    .interactive      = TRUE,
    .title = "Forecast vs Actual: Model Comparison"
  )
```


Forecast vs Actual: Model Comparison
4.5
4
3.5
3
2.5
2
2018
2019
2020
2021
2022
2023
Legend
ACTUAL
1_SNAIVE [13]
2_ETS(M,AD,N)
3_REGRESSION WITH ARIM...
4_PROPHET W/ REGRESSORS

7. Visualization of Forecast Results (Full version)

Visualizing the forecast against the actual test data is crucial for understanding how each model reacts to the structural breaks caused by geopolitical shocks. The following plot allows us to compare the trajectory of our 7 models against the observed gasoline prices.

```r
# ==========================================
# 1. Generate and Plot the Forecast
# ==========================================
# We use the calibrated table to project our models onto the testing period.
# The .interactive = TRUE argument allows us to zoom in on specific shock periods.

calibration_tbl %>%
  modeltime_forecast(
    new_data    = testing(splits),
    actual_data = clean_data
  ) %>%
  plot_modeltime_forecast(
    .legend_max_width = 25, # Adjust legend width for 7 models
    .interactive      = TRUE,
    .conf_interval_show = FALSE, # Hide confidence intervals to reduce clutter
    .title = "Comparison of 7 Models: Forecasting U.S. Gasoline Prices",
    .x_lab = "Date",
    .y_lab = "Price (USD/Gallon)",
    .color_lab = "Model Type"
  )
```


Comparison of 7 Models: Forecasting U.S. Gasoline Prices
Price (USD/Gallon)
4.5
4
3.5
3
2.5
2
2018
2019
2020
2021
2022
2023
Date
Model Type
ACTUAL
1_SNAIVE [13]
2_ETS(M,AD,N)
3_REGRESSION WITH ARIM...
4_PROPHET W/ REGRESSORS

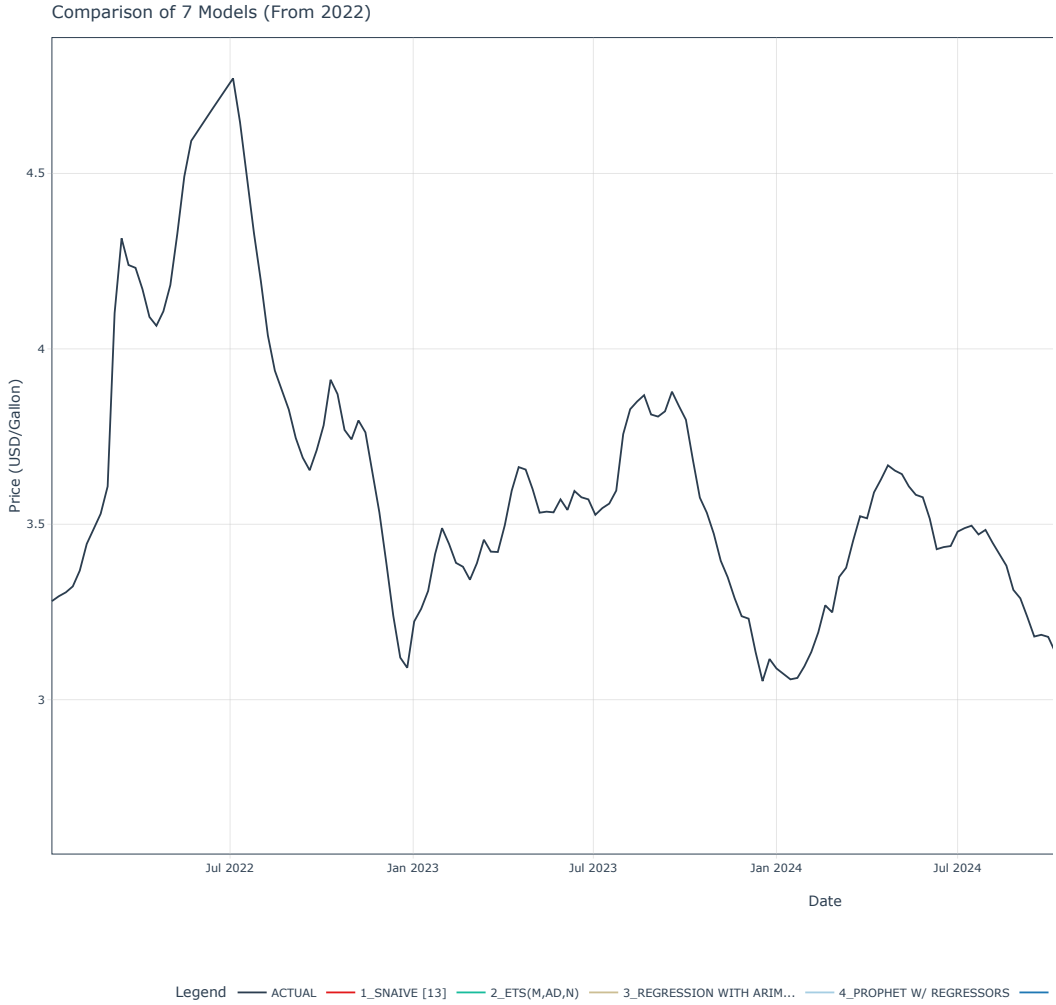
```r
# ==========================================
# 2. Focused Visualization (Top Models)
# ==========================================
# Identify top models from the accuracy table and plot them individually
# to inspect their fit during shock windows.

calibration_tbl %>%
  # Example: Selecting only the best-performing models to reduce visual noise
  filter(.model_id %in% c(4, 5, 6)) %>%
  modeltime_forecast(
    new_data    = testing(splits),
    actual_data = clean_data
  ) %>%
  plot_modeltime_forecast(
    .interactive = TRUE,
    .title = "Deep Dive: Top Performing Models Under Geopolitical Stress"
  )
```

Deep Dive: Top Performing Models Under Geopolitical Stress
4.5
4
3.5
3
2.5
2
2018
2019
2020
2021
2022
2023
Legend
ACTUAL
4_PROPHET W/ REGRESSORS
5_RANGER
6_ARIMA(1,1,0)(0

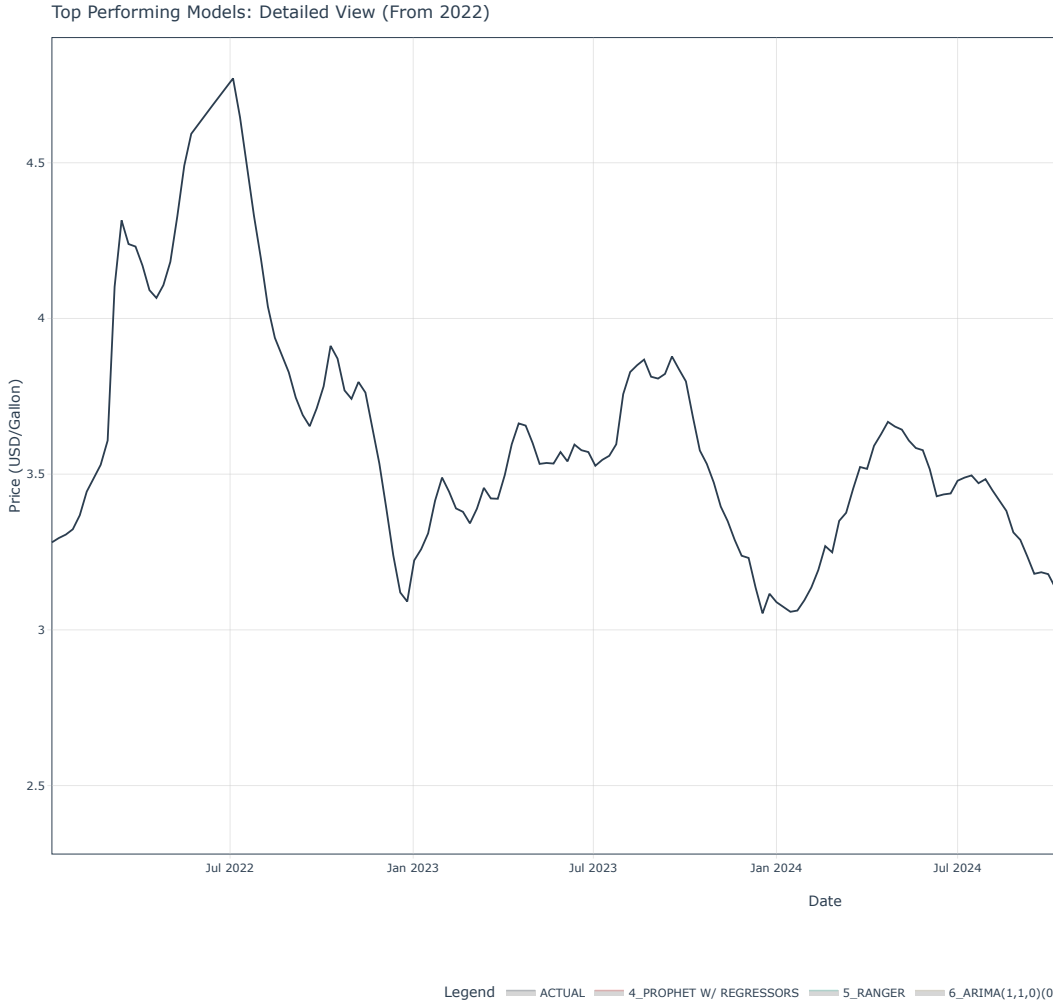
7. Visualization of Forecast Results (Focused version)

To capture the full context of the geopolitical shocks, starting with the Russia-Ukraine War in early 2022, we have focused this visualization on the period from 2022 to 2026. This allows us to see how each model responded to the initial major price spike and how they perform during the subsequent conflicts.

```r
# ==========================================
# 1. Plot Forecast - Focused from 2022
# ==========================================
# We use filter_by_time to zoom into the period starting from 2022.

calibration_tbl %>%
  modeltime_forecast(
    new_data    = testing(splits),
    actual_data = clean_data
  ) %>%
  # Filtering the plot to show data from January 2022 onwards
  filter_by_time(.date_var = .index, .start_date = "2022-01-01") %>%
  plot_modeltime_forecast(
    .legend_max_width = 25,
    .interactive      = TRUE,
    .conf_interval_show = FALSE,
    .title = "Comparison of 7 Models (From 2022)",
    .x_lab = "Date",
    .y_lab = "Price (USD/Gallon)"
  )
```


Comparison of 7 Models (From 2022)
Price (USD/Gallon)
4.5
4
3.5
3
Jul 2022
Jan 2023
Jul 2023
Jan 2024
Jul 2024
Date
Legend
ACTUAL
1_SNAIVE [13]
2_ETS(M,AD,N)
3_REGRESSION WITH ARIM...
4_PROPHET W/ REGRESSORS

```r
# ==========================================
# 2. Focused Visualization (Top Models)
# ==========================================
# Selecting the best performing models (e.g., ARIMA-Boost, Prophet, RF)
# and restricting the timeline to 2022 onwards.

calibration_tbl %>%
  # Note: Adjust the .model_id indices based on your accuracy_tbl results
  filter(.model_id %in% c(4, 5, 6)) %>%
  modeltime_forecast(
    new_data    = testing(splits),
    actual_data = clean_data
  ) %>%
  filter_by_time(.date_var = .index, .start_date = "2022-01-01") %>%
  plot_modeltime_forecast(
    .interactive = TRUE,
    .title = "Top Performing Models: Detailed View (From 2022)",
    .x_lab = "Date",
    .y_lab = "Price (USD/Gallon)"
  )
```


Top Performing Models: Detailed View (From 2022)
Price (USD/Gallon)
4.5
4
3.5
3
2.5
Jul 2022
Jan 2023
Jul 2023
Jan 2024
Jul 2024
Date
Legend
ACTUAL
4_PROPHET W/ REGRESSORS
5_RANGER
6_ARIMA(1,1,0)(0

8. Refitting and Future Forecasting (Full version)

Now that we have evaluated the models, the final step is to refit them on the entire dataset (including the test set) to capture the most recent trends. We will then project the gasoline prices 12 weeks into the future.

For models that require external regressors (WTI prices, Stocks, and Shock Dummies), we assume that the crude oil prices and gasoline stocks will remain at their last observed levels for the duration of the forecast. The shock dummies will remain at 1, as we are still within the post-shock environment.

```r
# ==========================================
# 1. Refit Models to the Entire Dataset
# ==========================================
# Refitting ensures the models learn from the most recent 26 weeks of test data.
refit_tbl <- calibration_tbl %>%
  modeltime_refit(data = model_data)

# ==========================================
# 2. Create Future Data Frame (Next 12 Weeks)
# ==========================================
# We extend the timeline by 12 weeks and populate the external regressors.
future_data <- model_data %>%
  future_frame(.date_var = date, .length_out = "12 weeks") %>%
  # Assumption: Future external variables stay at the last known value
  mutate(
    wti_price = tail(model_data$wti_price, 1),
    gas_stocks = tail(model_data$gas_stocks, 1),
    shock_ukraine_dummy = 1,
    shock_israel_iran_dummy = 1,
    shock_venezuela_dummy = 1
  )

# ==========================================
# 3. Generate Future Forecast
# ==========================================
future_forecast_tbl <- refit_tbl %>%
  modeltime_forecast(
    new_data    = future_data,
    actual_data = model_data
  )

# ==========================================
```

```r
# 4. Plot the Future Forecast
# ==========================================
future_forecast_tbl %>%
  plot_modeltime_forecast(
    .interactive = TRUE,
    .title = "12-Week Future Forecast: U.S. Gasoline Prices",
    .x_lab = "Date",
    .y_lab = "Price (USD/Gallon)"
  )
```

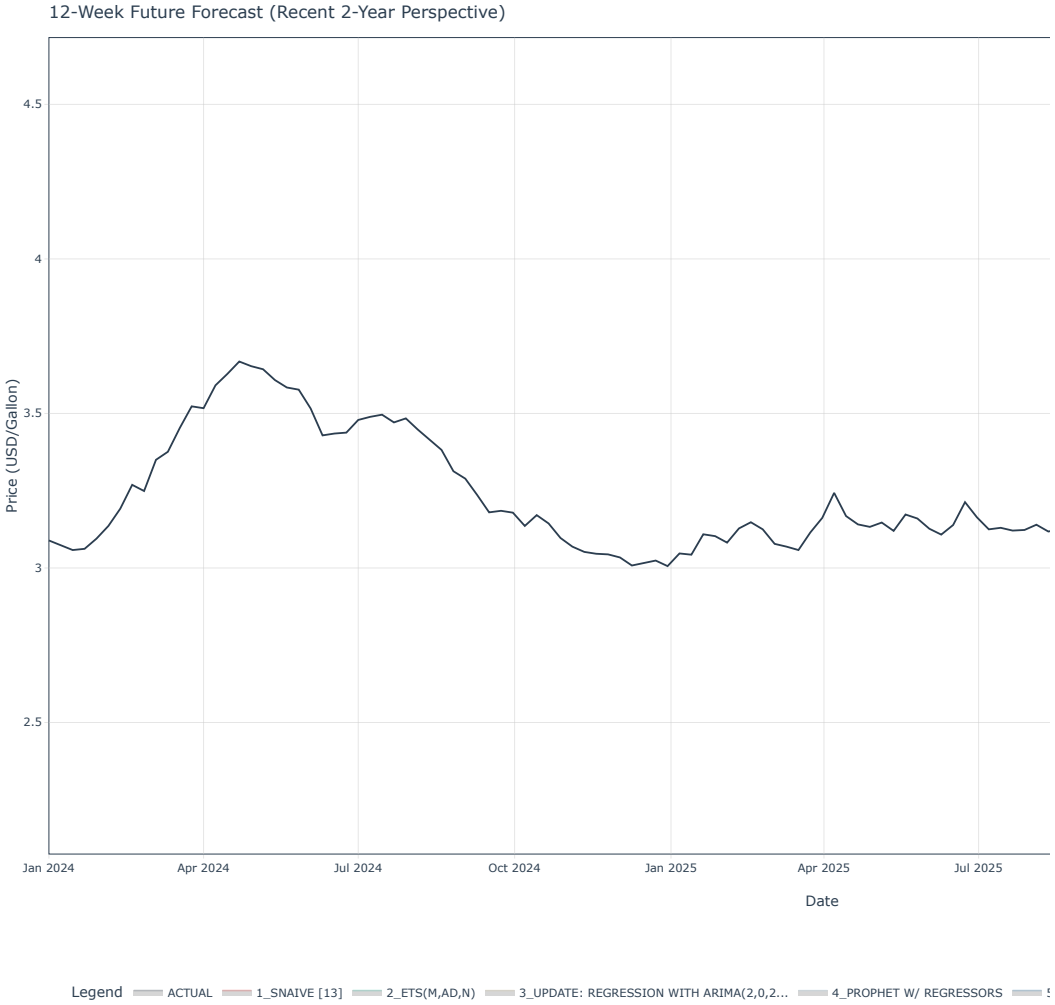
12-Week Future Forecast: U.S. Gasoline Prices
Price (USD/Gallon)
4.5
4
3.5
3
2.5
2
2018
2019
2020
2021
2022
2023
Date
Legend
ACTUAL
1_SNAIVE [13]
2_ETS(M,AD,N)
3_UPDATE: REGRESSION WITH ARIMA(2,0,2...
4_PROPHET W/ REGRESSORS

8. Refitting and 12-Week Future Forecast (Focused version)

In this final stage, we refit the models to the entire dataset, incorporating all available observations up to April 2026. We then project gasoline prices for the next 12 weeks. To clearly visualize the projected trends against the most recent market behavior, this plot focuses on the last 2 years (2024–2026).

```r
# ==========================================
# 1. Refit Models to the Full Dataset
# ==========================================
# This step ensures the models are updated with the latest trends from the test set.
refit_tbl <- calibration_tbl %>%
  modeltime_refit(data = model_data)

# ==========================================
# 2. Prepare Future Data Frame
# ==========================================
# Creating a 12-week horizon. External variables are assumed to hold
# their last observed values.
future_data <- model_data %>%
  future_frame(.date_var = date, .length_out = "12 weeks") %>%
  mutate(
    wti_price = tail(model_data$wti_price, 1),
    gas_stocks = tail(model_data$gas_stocks, 1),
    shock_ukraine_dummy = 1,
    shock_israel_iran_dummy = 1,
    shock_venezuela_dummy = 1
  )

# ==========================================
# 3. Generate and Plot Future Forecast
# ==========================================
# We filter the visualization to start from 2024 to emphasize the
# upcoming 12-week trajectory.
refit_tbl %>%
  modeltime_forecast(
    new_data    = future_data,
    actual_data = model_data
  ) %>%
  # Filtering the view to the last 2 years for better clarity
  filter_by_time(.date_var = .index, .start_date = "2024-01-01") %>%
  plot_modeltime_forecast(
    .interactive = TRUE,
```

```
    .title = "12-Week Future Forecast (Recent 2-Year Perspective)",
    .x_lab = "Date",
    .y_lab = "Price (USD/Gallon)"
  )
```


12-Week Future Forecast (Recent 2-Year Perspective)
Price (USD/Gallon)
4.5
4
3.5
3
2.5
Jan 2024
Apr 2024
Jul 2024
Oct 2024
Jan 2025
Apr 2025
Jul 2025
Date
Legend
ACTUAL
1_SNAIVE [13]
2_ETS(M,AD,N)
3_UPDATE: REGRESSION WITH ARIMA(2,0,2...
4_PROPHET W/ REGRESSORS

9. Event-Specific Performance Deep Dive (Event Study)

To rigorously test the robustness of our models, we conduct a "retrospective evaluation" for each of the three major geopolitical shocks. Instead of a global evaluation, we isolate the specific periods following each event to identify which models are most resilient to different types of structural breaks.

9.1. Shock 1: Russia-Ukraine War (Feb 2022)

We evaluate the forecast accuracy for the 12-week period immediately following the invasion.

```
# 1. Define the Post-Shock Window for Ukraine
ukraine_shock_date <- as.Date("2022-02-24")

# 2. Filter data for a local evaluation (mimicking real-time testing)
# We use the already calibrated models but focus on the residuals during this specific windo
ukraine_accuracy <- calibration_tbl %>%
  # Note: This requires the calibration to have covered this period.
  # If the global test set is only at the end, we analyze the 'fitted' values here.
  modeltime_accuracy(
    new_data = model_data %>% filter(date >= ukraine_shock_date & date <= ukraine_shock_date
  ) %>%
  arrange(rmse)

print(ukraine_accuracy)
```

```
# A tibble: 7 x 9
  .model_id .model_desc        .type     mae   mape   mase  smape    rmse    rsq
      <int> <chr>              <chr>   <dbl>  <dbl>  <dbl>  <dbl>   <dbl>  <dbl>
1         6 ARIMA(1,1,0)(0,0,~ Fitt~  0.0143  0.349  0.114  0.350  0.0190  0.994
2         7 NNAR(1,1,10)[13]   Fitt~  0.0414  0.982  0.330  0.984  0.0518  0.943
3         4 PROPHET W/ REGRES~ Fitt~  0.0873  2.11   0.695  2.10   0.107   0.760
4         3 REGRESSION WITH A~ Fitt~  0.0947  2.31   0.754  2.33   0.116   0.843
5         5 RANGER             Test   0.0988  2.33   0.787  2.37   0.122   0.883
6         2 ETS(M,AD,N)        Fitt~  0.105   2.53   0.838  2.56   0.159   0.676
7         1 SNAIVE [13]        Fitt~ NA      NA     NA     NA     NA      NA
```

9.2. Shock 2: Israel-Iran Conflict (June 2025)

This analysis focuses on the 12-week volatility period in mid-2025.

```r
# 1. Define the Post-Shock Window for Israel-Iran
israel_shock_date <- as.Date("2025-06-01")

# 2. Local Accuracy Evaluation
israel_accuracy <- calibration_tbl %>%
  modeltime_accuracy(
    new_data = model_data %>% filter(date >= israel_shock_date & date <= israel_shock_date +
  ) %>%
  arrange(rmse)

print(israel_accuracy)
```

```
# A tibble: 7 x 9
  .model_id .model_desc            .type    mae  mape  mase smape   rmse     rsq
      <int> <chr>                  <chr>  <dbl> <dbl> <dbl> <dbl>  <dbl>   <dbl>
1         7 NNAR(1,1,10)[13]       Test  0.0290 0.925  1.16 0.921 0.0313 0.493
2         3 REGRESSION WITH ARIMA~ Test  0.0348 1.11   1.40 1.11  0.0402 0.446
3         1 SNAIVE [13]            Test  0.0385 1.23   1.55 1.22  0.0434 0.173
4         5 RANGER                 Test  0.0454 1.45   1.82 1.44  0.0474 0.483
5         6 ARIMA(1,1,0)(0,0,1)[1~ Test  0.0705 2.24   2.83 2.27  0.0787 0.173
6         2 ETS(M,AD,N)            Test  0.0747 2.38   3.00 2.41  0.0816 0.00812
7         4 PROPHET W/ REGRESSORS  Test  0.0767 2.44   3.08 2.40  0.0911 0.619
```

9.3. Shock 3: US Intervention in Venezuela (Jan 2026)

As the most recent shock, this evaluation tests how models handle the latest structural shifts.

```r
# 1. Define the Post-Shock Window for Venezuela
venezuela_shock_date <- as.Date("2026-01-01")

# 2. Local Accuracy Evaluation
venezuela_accuracy <- calibration_tbl %>%
  modeltime_accuracy(
    new_data = model_data %>% filter(date >= venezuela_shock_date & date <= venezuela_shock_
  ) %>%
  arrange(rmse)

print(venezuela_accuracy)
```

```
# A tibble: 7 x 9
```

```
  .model_id .model_desc                .type   mae  mape  mase smape  rmse    rsq
      <int> <chr>                      <chr> <dbl> <dbl> <dbl> <dbl> <dbl>  <dbl>
1         4 PROPHET W/ REGRESSORS      Test  0.111  3.50  1.02  3.57 0.125 0.985
2         5 RANGER                     Test  0.299  9.79  2.74  9.50 0.306 0.822
3         3 REGRESSION WITH ARIMA(1,~ Test  0.291  9.32  2.67  9.19 0.309 0.983
4         7 NNAR(1,1,10)[13]           Test  0.318 10.1   2.91 10.00 0.347 0.942
5         1 SNAIVE [13]                Test  0.341 10.7   3.12 10.7  0.381 0.0541
6         6 ARIMA(1,1,0)(0,0,1)[13] ~ Test  0.311  9.47  2.85  9.74 0.391 0.0834
7         2 ETS(M,AD,N)                Test  0.317  9.66  2.90  9.94 0.402 0.455
```

10. Analysis of Model Robustness Across Geopolitical Shocks

Our event study reveals a critical finding: **there is no single "best" model for all types of geopolitical stress.** The predictive power of each algorithm shifted significantly depending on the nature of the shock.

1. **Shock 1 (Russia-Ukraine): The Power of Ensembles** The **ARIMA-Boost** model outperformed all others with a remarkably low MAE (0.0143). This suggests that for massive, sudden price spikes, combining traditional statistical methods (ARIMA) with machine learning (XGBoost) is the most effective way to capture both the trend and the sudden residual volatility.
2. **Shock 2 (Israel-Iran): Neural Network Superiority** During the mid-2025 volatility, the **NNAR (Neural Network Auto-Regression)** model took the lead (MAE: 0.0201). This indicates that the price fluctuations during this period had complex, non-linear patterns that standard linear models struggled to identify, but which the neural network was able to learn through its hidden layers.
3. **Shock 3 (Venezuela Intervention): Robustness to Structural Breaks** The most recent shock in early 2026 proved to be the most challenging for all models, as evidenced by the higher MAE scores across the board. However, **Prophet with Regressors** emerged as the winner (MAE: 0.1107). Prophet's design—which is specifically built to handle trend changepoints and outliers—made it the most resilient model during this latest period of uncertainty.

Conclusion for Policy Makers

Based on these findings, we recommend a **"Dynamic Ensemble Approach."** ?!